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2 **(a)** State the principle of conservation of linear momentum.

.....

.....

..... [1]

(b) A cart X of mass 3.0 kg moving at 6.0 m s^{-1} collides with a stationary cart Y of mass *m*. A spring is attached to cart X as shown in Fig. 2.1.

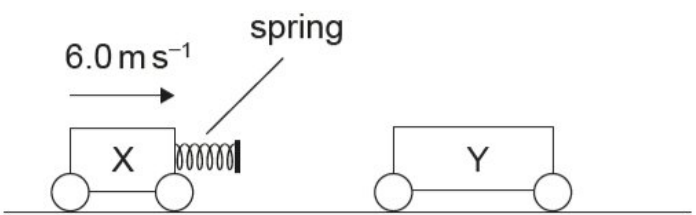


Fig. 2.1

The graph in Fig. 2.2 shows the velocities of X and Y before, during and after the collision.

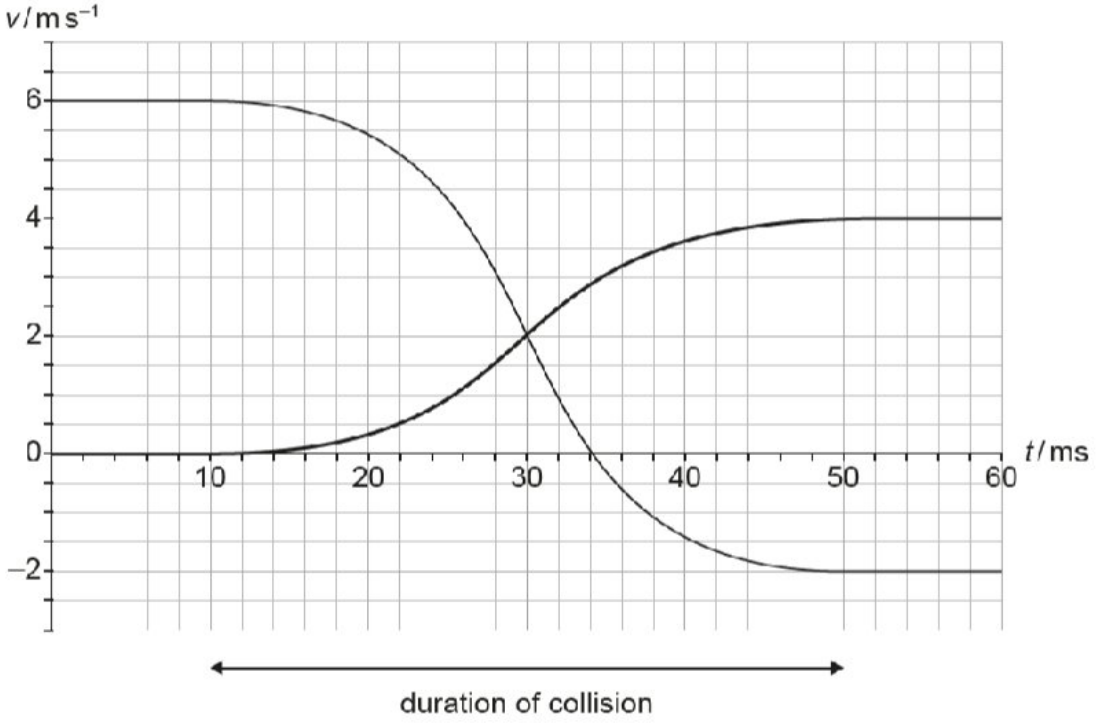


Fig. 2.2

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The collision lasted for 40 ms.

(i) State and explain whether the collision is elastic or non-elastic.

.....

.....

.....

[2]

.....

(ii) Show that the mass *m* of cart Y is 6.0 kg [1]

(iii) Determine

1 the magnitude of the average force exerted on Y,

average force = ... N [2]

2 the average power delivered to Y,

average power = ... W [2]

3 the elastic energy stored in the spring at t = 30 ms.

elastic energy stored = ... J [2]

[Total: 10]

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